Tectonophysics, 201 (1992) 83–96
Elsevier Science Publishers B.V., Amsterdam

Geodynamical implications of a Moscovian palaeomagnetic pole from the stable Saharan craton (Illizi basin, Algeria)

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(Received July 3, 1990; revised version accepted November 20, 1990)

ABSTRACT

Henry, B., Merabet, N., Yelles, A., Derder, M.M. and Daly, L., 1992. Geodynamical implications of a Moscovian palaeomagnetic pole from the stable Saharan craton (Illizi basin, Algeria). In: H. Perroud and G. Bylund (Editors), Palaeomagnetism of Old European Rocks. *Tectonophysics*, 201 (spec. sect.): 83–96.

Four different characteristic remanent magnetization (ChRM) orientations have been isolated in the El Adeb Larache formation of the Algerian Sahara (27.45° N, 8.9° E). Two of them are interpreted as overprints developed respectively during recent and Permian times. The two others give new Moscovian (28.7° S, 55.8° E) and probably Stephano–Autunian (38.5° S, 57.5° E) palaeopoles, in good agreement with those previously obtained from the Sahara, but significantly different from the Dwyka "varves" pole (McElhinny and Opdyke, 1968). Comparison with Lower Carboniferous results shows that the latitudinal displacement of Africa during the Carboniferous is mainly related to a clockwise rotation (about 25°) around an axis close to the northeastern border of Africa. Such a motion of Africa agrees well with geological observations and with the Pangea A2 hypothesis.

Introduction

Even if sound post-Jurassic palaeocontinental reconstructions have been obtained combining magnetic anomalies and palaeomagnetic data (Le Pichon and Fox, 1971; Pitman and Talwani, 1972; Olivet et al., 1984), evolution of the Pangea configuration from Carboniferous to Triassic is always hypothetical. Based mainly on palaeomagnetic data, several different evolution models have been proposed (Irving, 1977; Van der Voo, 1981; Merabet, 1987). One of the major uncertainties for this period is that the African apparent polar wander path (APWP) is poorly constrained (see Bachtadse and Briden, 1989, 1990; Feinberg et al., 1990) due to large time gaps in the palaeomagnetic data set as well as conflicting results for the same period (McElhinny and Opdyke, 1968; Daly and Irving, 1983).

Such incoherence in the results may be related to several causes. Firstly, the fact that some natural remanent magnetizations (NRM) are complete overprints, is not always easy to point out (see for example Triassic formations in Lybia and southern Tunisia—Nairn and Martin, 1978, and Ghorabi, 1990; the Gneiguira supergroup—Kent et al., 1984; or the Karoo basin—Ballard et al., 1986). Secondly, a lack of adequate demagnetization experiments in some studies may have masked possible multicomponent NRM's. Finally, age determination of rocks studied was sometimes doubtful. For reasons such as the last two mentioned, the Msissi norite results (Hailwood, 1974) can no longer be thought of as being representative of a Devonian palaeomagnetic pole (Salmon et al., 1987).

The conflicting results for the Upper Carboniferous (McElhinny and Opdyke, 1968—used as a reference for the Upper Carboniferous by Bachtadse and Briden, 1990—and Daly and Irving, 1983) clearly illustrate uncertainties in the determination of the African APWP. All these reasons

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greatly reduce the number of reliable palaeomagnetic determinations from Africa. This situation demands that additional palaeomagnetic investigations be undertaken, wherever possible within well-dated geological sections in a stable area such as the Sahara craton. In the Illizi basin, the well-dated Moscovian El Adeb Larache formation seems to be particularly favourable for such a study.

Geological setting

A good cross-section of the El Adeb Larache formation can be observed along the In Amenas–Illizi road between the Bourharet erg and the El Adeb Larache SONATRACH life base (Figs. 1 and 2). Here, outcrops are mainly dolomitic limestones with interbedded marls, sandstones and gypsum horizons. The limestones include some fauna-rich levels containing goniatites, brachiopods, gastropods and fusulinidae (Durif, 1959), clearly indicating a Moscovian (Upper Carboniferous: 315–296 Ma; Harland et al., 1982) age (Legrand-Blain, 1980). However, the top of the series outcropping near the Bourharet erg presents azoic facies rather similar to those of the Lower Tiguentourine formation, of Stephano–Autunian (296–268 Ma; Harland et al., 1982; Odin and Odin, 1990) age (Attar et al., 1981).

From a structural point of view, the Illizi basin appears to have been a relatively stable area since the Upper Carboniferous (see Fabre, 1988). The El Adeb Larache limestones are an almost horizontal formation (locally, northern to north-eastern dip lower than 4°). The different post-

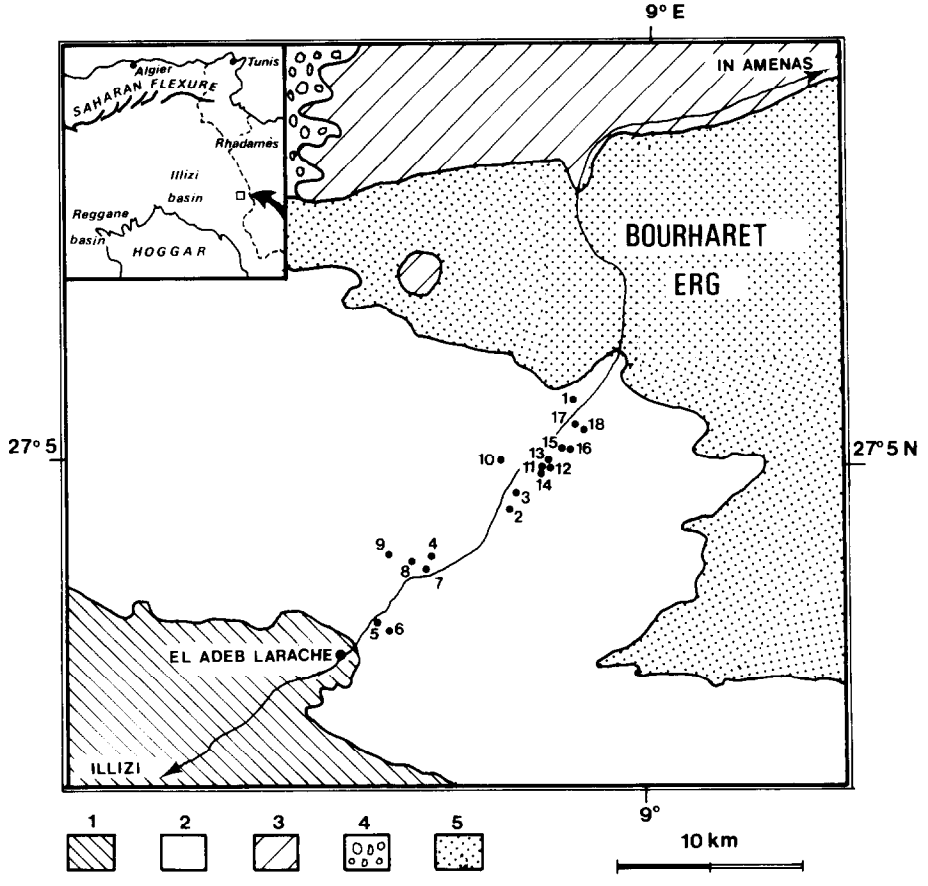


Fig. 1. Geological map of the studied area and sampling sites: pre-Moscovian formations (*1*), Moscovian (*2*), Lower Tiguentourine (*3*), Quaternary (*4*) and Erg (*5*). After Bonnet et al. (1965), Legrand-Blain (1980) and a SONATRACH internal report (in Attar et al., 1980).

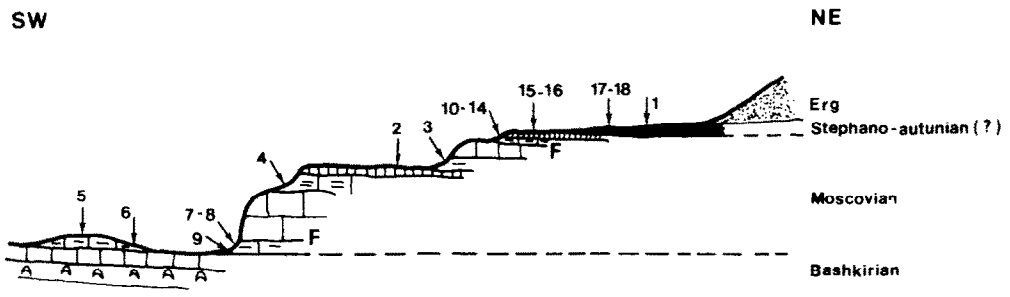


Fig. 2. NE–SW cross-section of the studied area (F = fossil-bearing levels) and sampling-site levels. The samples' lithology corresponds to fine dolomitic limestones, except in sites 1, 17, 18 (marls), 6 to 8 (porous dolomitic limestones) and some horizons of sites 3, 10 and 13 (marls and sandstones interbedded).

Paleozoic tectonic phases have so slightly affected the eastern part of the Algerian Sahara, that there is no evidence of deformation effects or volcanic activity in the studied area (the closest traces of recent volcanism being 100 km south-eastwards—Bossières and Megartsi, 1982). Although it was not possible to perform any of the standard palaeomagnetic stability tests used to date the acquisition of magnetization components (i.e. fold test, baked contact test etc.), this lack of tectonic or thermal events enhances the possibility of obtaining a primary magnetization direction.

Sampling and analysis procedures

178 oriented samples were collected at 18 sites distributed throughout the different stratigraphical levels (Fig. 2), including the fossil-bearing ones, and including 3 sites (34 samples) at the top of the series supposed to be of Stephano–Autunian age. At least seven cores were drilled at each site. One to three specimens have been obtained from each core.

Prior to demagnetization analysis, samples were stored in a zero field for at least one month to allow for a decay of the possible components of viscous remanent magnetization (VRM). Alternating field and thermal treatments were employed to characterize the magnetization components. Remanent magnetizations were measured using a JR-4 spinner magnetometer (GEOFYZIKA, Brno). Two devices, constructed at the Saint-Maur laboratory, were utilized for specimen demagnetization. The first is a cylindrical furnace with a fast heating level (reaching this level in 15 min and maintaining it for 45 min) in which the field is cancelled by four layers of Mu-metal which are demagnetized before each treatment. The specimens were cooled for one hour by forced air at the heating site. The second apparatus, used for alternating field demagnetization, automatically cancels the parasitic anhysteretic effect (Le Goff, 1985) up to 280 mT. Two translation inductometers, also constructed in the laboratory, allowed the determination of thermomagnetic curves using a field of 5 or 10 mT according to sample, and of a hysteresis loop with an electromagnet giving a field up to 1.6 T.

Thermal demagnetization was performed in steps ranging from 100 °C to 30 °C. Reduced temperature intervals were employed at the highest temperatures in order to permit a more detailed study of the remanence evolution. To evaluate mineralogical change during thermal treatment, low field magnetic susceptibility was measured at room temperature after each heating stage using a KLY-2 Kappabridge (GEOFYZIKA, Brno).

Data analysis was carried out using classical methods: (1) remaining magnetization after each treatment (remaining vector); (2) magnetization lost between each treatment (vector difference); (3) use of remagnetization circles (Halls, 1978; Kirschvink, 1980); and (4) presentation of data on equal area projection diagrams and on orthogonal vector plots (Wilson and Everitt, 1963; Zijderveld, 1967). Mean directions of the stable magnetization component were computed using Fisher statistics (1953).

Rock- and palaeomagnetic measurements

Three groups of different NRM intensities can be distinguished. The high-intensity one (ranging about a mean value of 4×10^{-3} A/m) corresponds mainly to relatively porous sandstones with hematite-rich thin layers, or to orange marls. The low-intensity ones (ranging mainly from about $3 \times$ to 20×10^{-5} A/m) consist of homogeneous fine limestones and marls.

Curie point determinations show the existence of two magnetic minerals according to the samples. Surprisingly, magnetite was found to be the main magnetic mineral in a level of orange marls at site 3; the curves of induced magnetization versus temperature during heating and cooling in air are then relatively similar (Fig. 3a). Most specimens show a typical dominant effect of hematite, but many of them have a cooling curve very different from the heating one (Fig. 4a), indicating a decrease in the proportion of hematite and the formation of magnetite during

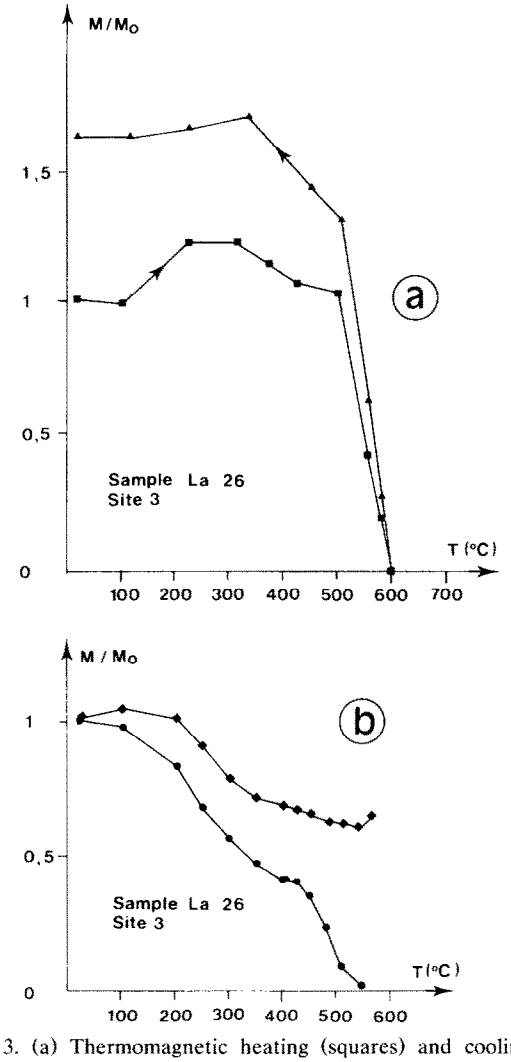


Fig. 3. (a) Thermomagnetic heating (squares) and cooling (triangles) curves. (b) Normalized magnetic susceptibility (diamonds) and NRM intensity (circles) at room temperature as a function of applied temperature; sample containing magnetite (La 26 from site 3).

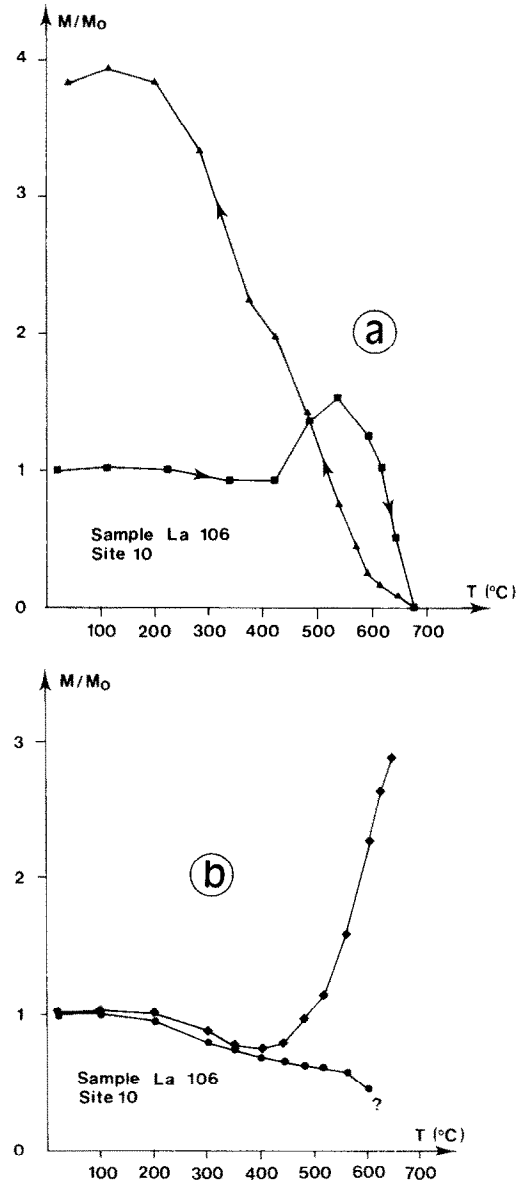


Fig. 4. See Fig. 3; sample containing hematite (La 106 from site 10).

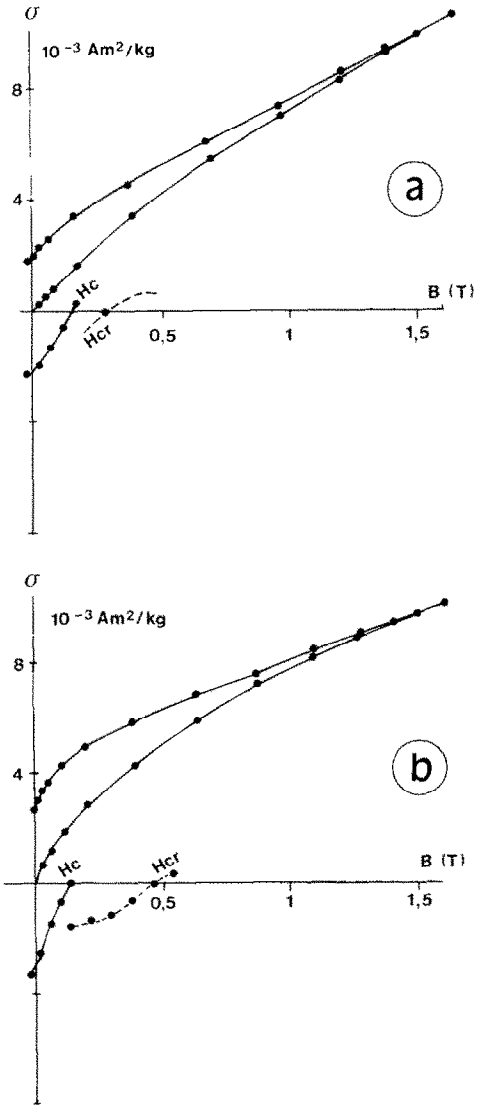


Fig. 5. Hysteresis loop before (a) and after (b) heating at 675 ° C of a sample (La 106 from site 10) with hematite as the main magnetic mineral before heating (*Hc* = coercive induction; *Hcr* = remanent coercive induction).

the thermal treatment in air. Since heating in air generally induces oxidation, these specimens probably contain some reducing components, perhaps related to the origin of magnetite formation at site 3 (see the discussion). Hysteresis loop analysis of the same specimens confirms the difference in magnetic mineralogy composition before and after heating (Fig. 5): unheated rock has higher coercive induction and lower saturation magnetization and remanent coercive induction than heated rock. Moreover, the latter presents a

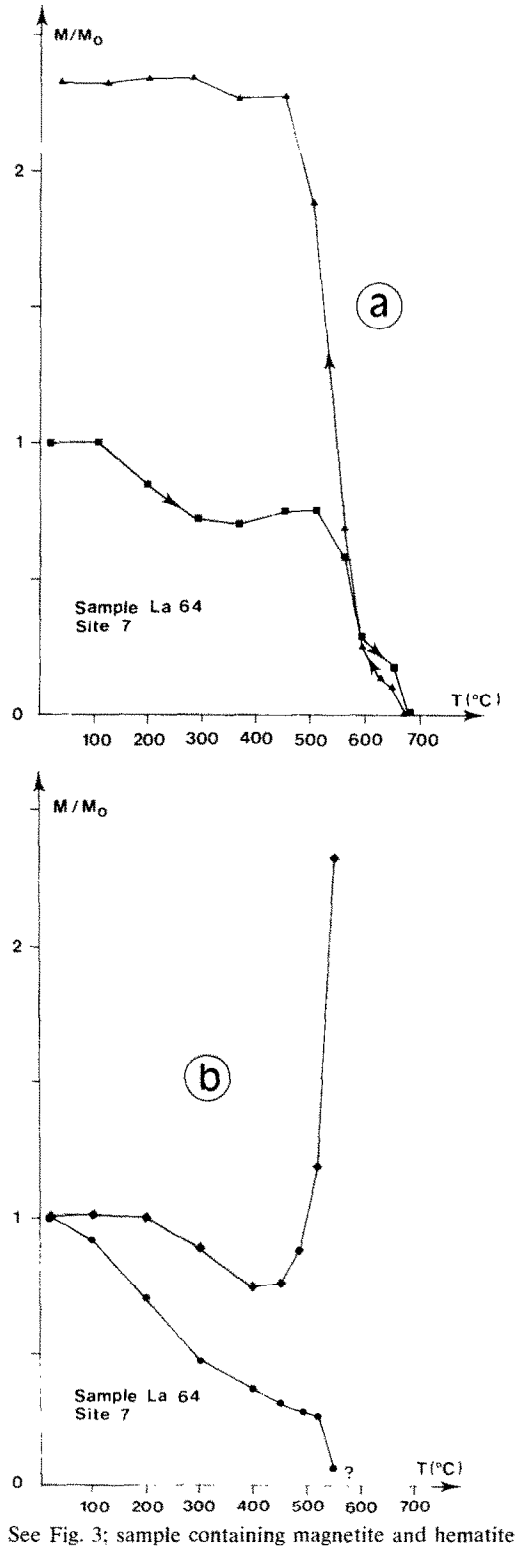


Fig. 6. See Fig. 3; sample containing magnetite and hematite (La 64 from site 7).

partial saturation superimposed on the paramagnetic–antiferromagnetic effect (which is then a little less important than for unheated specimens, probably because of the decreased proportion of hematite—see above). Finally (Fig. 6a), some specimens (from sites 6 to 8) show the presence of both magnetite and hematite carriers. The relative stability (see below), up to 500 °C, of the susceptibility measured at room temperature after each heating stage of the thermal analysis shows that magnetite does not appear during heating at these temperatures, and is thus already present in the specimens before heating.

Most specimens show a slight decrease in magnetic susceptibility at room temperature after thermal treatment at the 200 or 300 °C heating stage (Figs. 3b, 4b and 6b). Since remanent magnetization at these stages is mostly in a stable direction and does not present significant change in the intensity decrease and in magnetic viscosity, this slight susceptibility decrease is probably related to chemical changes in grains which are not the main carriers of the remanent magnetization. However, after higher temperature stages (> 500 °C) there appears, in samples containing hematite, a strong increase in susceptibility and magnetic viscosity, related to hematite reduction and affecting the grains which carry remanent magnetization. Therefore the main part of the data corresponding to the highest temperatures (from 550 to 630 °C heating, according to specimens) cannot be taken into account because of directional unstability of the magnetization above these temperatures.

The NRM vectors are displayed in a girdle of directions between that of the present Earth's magnetic field at the site and SSE horizontal (Fig. 7); this distribution reflects the existence of at least two components, one of them being possibly of a viscous origin.

Alternating field treatment generally allows only partial demagnetization due to the presence of a high-coercivity phase, except in samples with magnetite as the main ferrimagnetic component, for which thermal and alternating field treatments give a quite similar characteristic remanent magnetization (ChRM) direction. Thermal treatment has therefore been used for all the samples as it allows more complete characterization of the NRM.

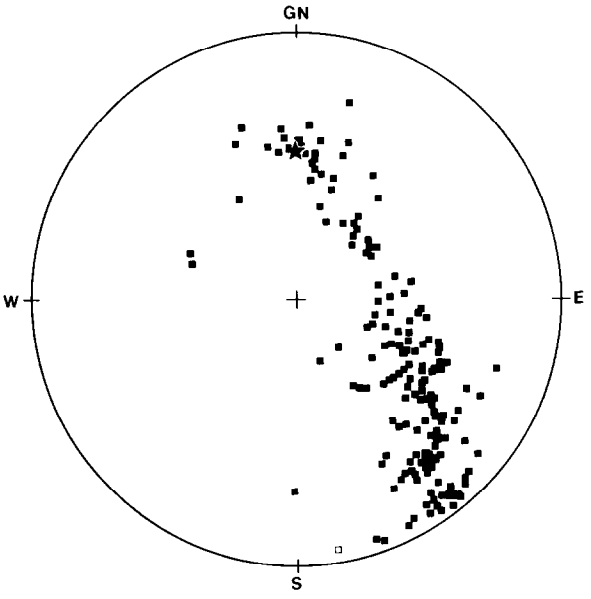


Fig. 7. Equal area plot (filled symbols: lower hemisphere; open symbols: upper hemisphere) of NRM vectors before dip correction. Star indicates the present Earth's magnetic field direction.

During thermal treatments, two kinds of magnetization evolution are recognized. The first is characterized by the existence of two or more components with superimposed unblocking temperature spectra: magnetization of the sample does not reach a stable direction, even after the highest significant thermal steps (e.g. sites 6 to 8). The second one shows either two magnetization components with separate unblocking temperature spectra (Fig. 8c), or a single component with a thermally distributed unblocking spectrum (Figs. 8a, 8b and 8d). When present, the low-temperature component (destroyed at temperatures lower than 200 °C) is directionally similar to the present Earth's magnetic field and is thus inferred to be of a recent viscous origin. The stable high temperature component directions are shown in Fig. 9.

Discussion

Superimposed magnetization components

After analysis, three sites (6 to 8), corresponding to neighbouring levels of porous granular

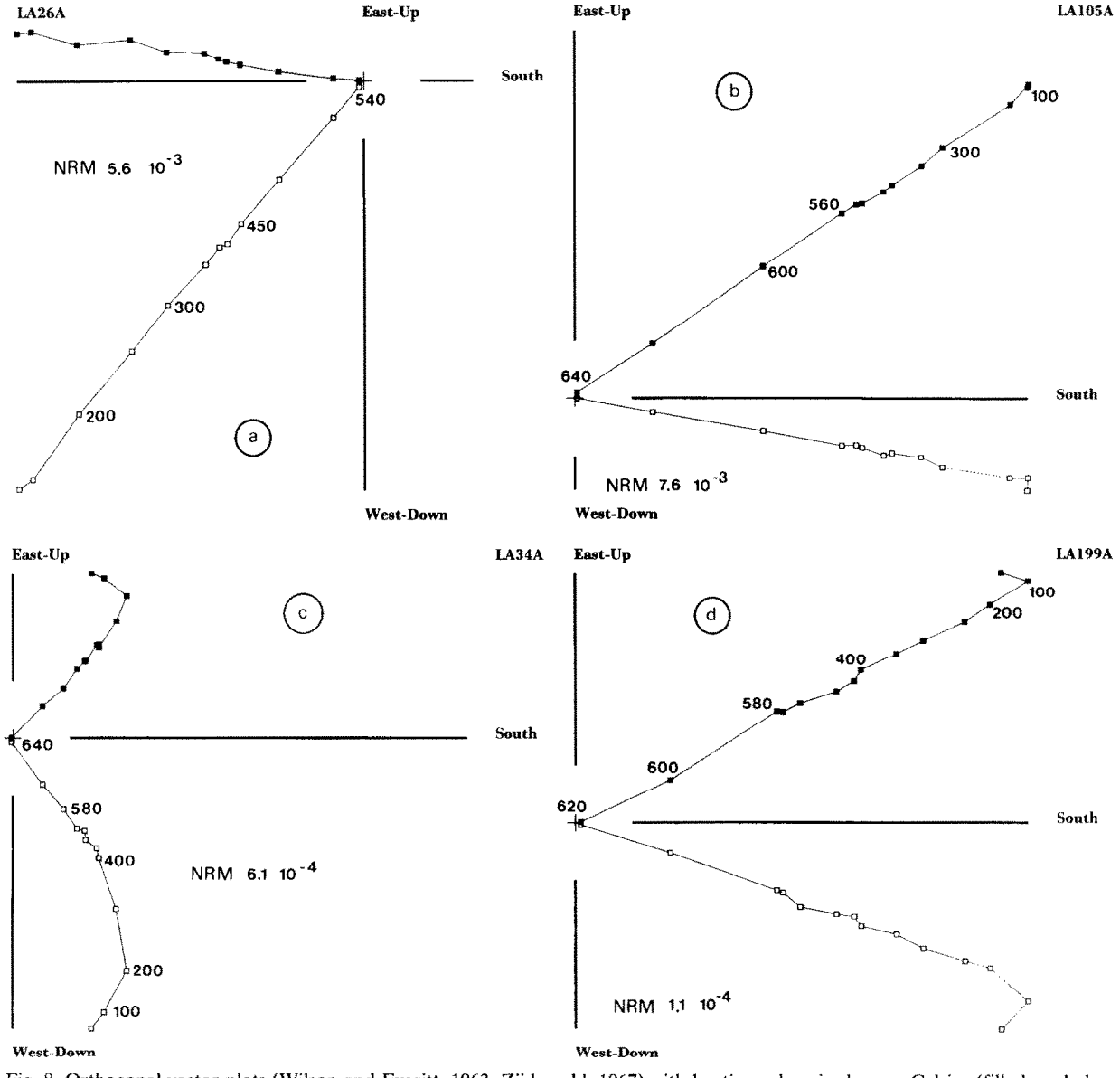


Fig. 8. Orthogonal vector plots (Wilson and Everitt, 1963; Zijderveld, 1967) with heating values in degrees Celsius (filled symbols: horizontal plane; open symbols: vertical plane) of specimens after dip correction, where ChRM has been isolated (NRM in A/m): (a) recent overprint (specimen La 26a from site 3); (b) Permian overprint (specimen La 105a from site 10); (c) low blocking recent component and Moscovian magnetization (specimen La 34a from site 4); (d) Stephano–Autunian magnetization (specimen La 199a from site 18).

limestones, are eliminated for directional statistics because their superimposed magnetizations could not be completely separated, except for five samples which show similar stable high temperature magnetization directions to that of the Permian overprint (see below and Table 1). A component corresponding to the lowest unblocking temperatures is well defined as a recent overprint by its vectorial differences rather close to the recent field. Remagnetization circles (Halls, 1978; Kirschvink, 1980) are mostly very close to one another (Fig. 10), but the best estimate of the main component, determined by the method of McFadden and McElhinny (1988) is $D = 143.0°$, $I = 10.4°$ ($\alpha_{95} = 1.6°$), close to the direction of the Permian overprint. Magnetization in samples

from sites 6 to 8 is therefore mainly related to superimposition of two overprints. It has been noticed that these samples are those containing both hematite and magnetite (but strong mineralogical changes after heating at 500 °C do not allow to obtain significant data for higher temperatures, except for the five specimens indicated above).

Single ChRM

At the remaining fourteen sites (samples from site 14 have a magnetization intensity too low to obtain precise results), the magnetic treatment allowed, for the most of specimens, the isolation of a single ChRM. This ChRM may have a different orientation according to the site or, some-

TABLE 1

Number of samples (N), mean direction (D, I) for each cluster of ChRM in each site after dip correction; mean values for each cluster and virtual geomagnetic pole (lat., long.) [a]

Site	N	D(°)	I(°)	k	α_{95}(°)	Lat. (°S)	Long. (°E)	K	A_{95}(°)
Recent overprint									
3	6	9.2	48.1	107	5.5				
10	5	7.4	44.0	1461	1.6				
Mean	11	7.3	46.2	176	3.2	−83.5	96.4		
Mean	2 sites	8.3	46.0	–	–	−82.6	97.6		
Permian overprint (?)									
7	2	140.0	5.0	–	–				
8	3	138.7	9.3	1167	2.3				
10	7	144.0	6.1	324	2.9				
13	10	144.5	6.5	247	2.8				
Mean	22	142.2	6.7	270	1.8	42.3	64.7		
Mean	4 sites	140.8	6.8	564	2.9	41.2	65.9		
Mean	4 sites	–	–	–	–	42.0	65.1	717	2.6
Primary directions									
Moscovian:									
2	8	141.1	40.3	220	3.3				
3	5	141.3	34.0	43	9.6				
4	8	138.9	42.8	85	5.4				
5	8	136.8	28.6	229	3.3				
9	5	136.0	29.6	436	3.0				
10	4	132.2	27.8	443	3.3				
11	9	133.7	28.8	148	3.8				
12	7	132.9	28.6	203	3.7				
15	2	144.0	37.0	–	–				
16	5	140.8	31.5	338	3.4				
Mean	61	136.1	33.1	83	2.0	27.7	57.0		
Mean	10 sites	137.6	33.0	161	3.5	28.7	55.9		
Mean	10 sites	–	–	–	–	28.7	55.8	235	2.9
Stephano–Autunian (?):									
1	11	142.0	21.4	107	4.1				
17	13	142.9	19.2	381	2.0				
18	10	145.4	16.9	278	2.7				
Mean	34	142.3	19.8	141	2.0	37.5	58.3		
Mean	3 sites	143.4	19.2	837	2.8	38.5	57.5		
Mean	3 sites	–	–	–	–	38.5	57.5	1459	2.1

[a] Corresponding to Fisher's (1953) parameters (k, α_{95}, K, A_{95}).

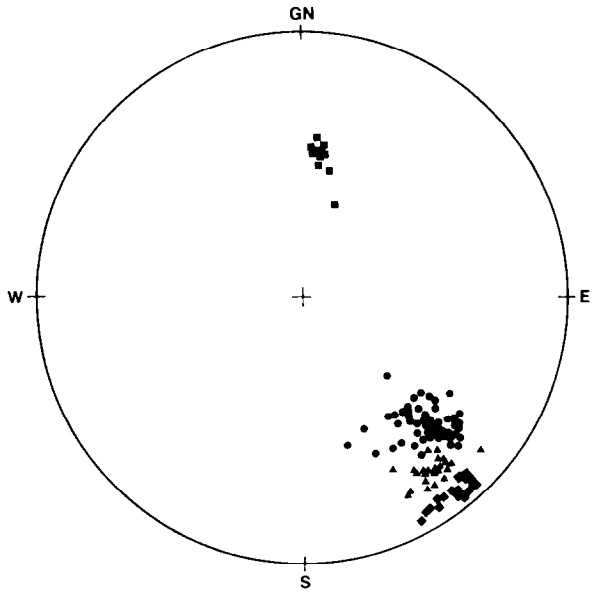


Fig. 9. Equal area plot (see Fig. 7) of ChRM after dip correction: recent overprint (squares); Permian overprint (diamonds); Moscovian directions (circles); Stephano–Autunian directions (triangles).

times, to each stratigraphical level at the same site (sites 3 and 10). Both the coherence of magnetization directions within each site (except in the particular case of these sites 3 and 10 where,

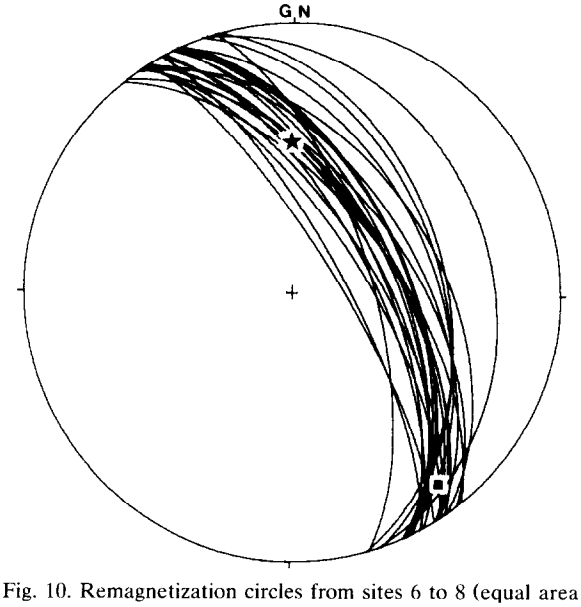


Fig. 10. Remagnetization circles from sites 6 to 8 (equal area plot, lower hemisphere). The square indicates the best estimate of the main component (McFadden and McElhinny, 1988). The star indicates the present Earth's magnetic field direction.

respectively, two and three significantly different clusters of magnetization direction are present—see below) and the significant difference of the mean direction between the studied sites (Table 1), clearly show that this magnetization does not result from overlapping of two or more magnetization components like in sites 6 to 8. The ChRM directions may be gathered into four main clusters after tectonic correction (Fig. 9 and Table 1). These clusters are now discussed in chronological order from the youngest to the oldest interpreted magnetization age.

Recent overprint

The first cluster, derived from 11 samples, has a mean direction ($N = 11$, $D = 7.3°$, $I = 46.2°$, $\alpha_{95} = 3.2°$ after dip correction; $D = 7.1°$, $I = 44.1°$, $\alpha_{95} = 3.2°$ before dip correction) close to the present Earth's field (Fig. 9). This magnetization carried by the whole unblocking temperature spectra (Figs. 3a and 8a) is therefore a complete overprint which has developed during relatively recent times. Samples which carry this remagnetization generally have a strong NRM intensity and include all those having magnetite as the main magnetization carrier. Similarly, as the partial recent overprint of site 6 to 8, an effect related to chemically active fluids (similar recent overprints, with normal or reversed polarity according to vertical elevation, observed in the same area within some underlying Upper Devonian–Middle Carboniferous formations, have been interpreted as related to level variation of such fluids—Henry et al., 1990) might be at the origin of this chemical remanent magnetization (CRM).

Permian overprint

The second cluster is defined by a low inclination towards a SSE declination ($N = 22$, $D = 142.2°$, $I = 6.7°$, $\alpha_{95} = 1.8°$). This ChRM (Fig. 8b) direction has been obtained at the bottom (sites 7–8) and at the top (in site 13 and locally in site 10) of the Moscovian formation, and in samples of relatively porous limestones and sandstones, often (sites 10 and 13) with thin hematite-rich layers. In this cluster, the NRM intensity is mostly of the same order as that of the samples with a recent overprint, but with

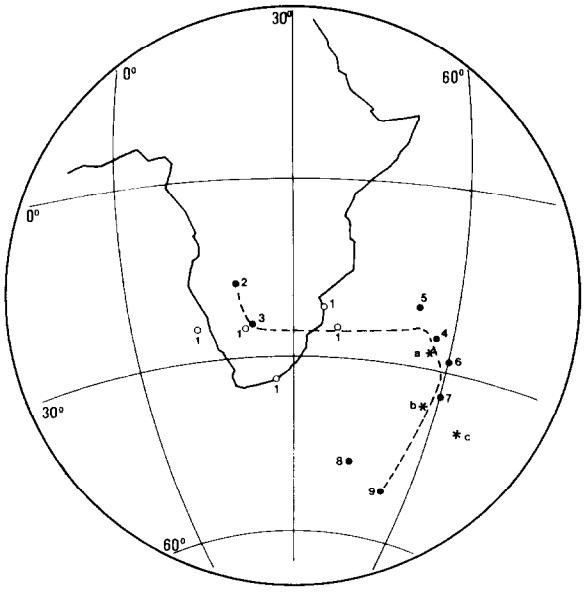


Fig. 11. Upper Devonian–Triassic VGPs and corresponding APWP for stable Africa. Circles: Dwyka sites (*1*) (McElhinny and Opdyke, 1968). Dots: *2* = Fammenian of Ben Zireg (Aifa et al., 1989); *3* = Tournaisian remagnetization at Ben Zireg (Aifa et al., 1989); *4* = Upper Namurian–Lower Moscovian of Hassi Bachir (Daly and Irving, 1983); *5* = Lower Moscovian of Ain Ech Chebbi (Daly and Irving, 1983); *6* = Stephano–Autunian of Abadla (Morel et al., 1981); *7* = Permian remagnetization at Hassi Bachir (Daly and Irving, 1983); *8* = Permian remagnetization at Benis-Abbes (Aifa, 1987); *9* = Triassic of southern Tunisia (Ghorabi, 1990). Asterisks (this study): *a* = Moscovian; *b* = Stephano–Autunian; *c* = Permian overprint.

hematite as the magnetization carrier. Direction of this cluster is very coherent, but significantly different from those from all the neighbouring levels studied. The presence of such a direction of ChRM, very similar at the bottom and at the top of the series but significantly different of all the other ChRM directions, suggests that this direction could be related to a same remagnetization. The corresponding virtual geomagnetic pole (VGP) obtained by giving unit weight to each sample is situated at 42.3 ° S, 64.7 ° E. This pole is moreover close to the Permian remagnetization poles (Fig. 11) obtained in the formations of Hassi Bachir (35.5 ° S, 60.0 ° E—Daly and Irving, 1983) and Beni-Abbes (49.5 ° S, 42.2 ° E—Aifa, 1987). Our findings lead us to assume that this second cluster of magnetizations is related to an overprint during Permian times. In such areas without important tectonic and thermal events, magnetic overprints are likely related to chemical phenomena (possibly favoured here by the high rock porosity?).

Stephano–Autunian(?) and Moscovian directions

The other directions have been obtained from rocks of well cemented varied facies (limestones, marls) which comprise fine grains and a low degree of weathering. Such rocks are less susceptible to chemical remagnetization, and the ChRM direction (coherent within each site, but different from one site to another) probably corresponds to that of the primary magnetization. Contrarily to previous cases, NRM carried by hematite has a low intensity. Triangles and circles in Fig. 9 show a relatively important scattering in inclination, but the samples have been collected at different Moscovian levels, including probably also at the top some from the Stephano–Autunian (Fig. 2). The directions obtained from these upper levels ($N = 34$, $D = 142.3°$, $I = 19.8$, $\alpha_{95} = 2.0°$) are significantly different from the directions of the older levels (and in an intermediate orientation between magnetization of these older levels and Permian overprint); there is probably a relatively important stratigraphic break under these upper levels, which is in good agreement with the attribution of sites 1, 17 and 18 (from geological observation) to a Stephano–Autunian age. Moreover, this relation between stratigraphy and magnetization direction here clearly confirms that the direction of magnetization from the third and fourth clusters is that of the primary magnetization.

The last and most important cluster, from well-dated Moscovian levels, is characterized by a higher value of inclination (Fig. 8c). It is defined by the following mean direction: $N = 61$, $D = 136.1°$, $I = 33.1°$, $\alpha_{95} = 2.0°$. This direction is not significantly different from that calculated by giving a unit weight to each of the ten sites ($D = 137.6°$, $I = 33.0°$, $\alpha_{95} = 3.5°$). The great number of levels studied in this Moscovian formation gives an important weight to this result for the African APWP.

Palaeopoles

We have summarized in Table 2 all the palaeopoles from Upper Devonian to Permian for stable Africa. Most of them have been obtained from stable Saharan craton formations. The only exception concerns the Dwyka pole from central Africa (McElhinny and Opdyke, 1968), used as an Upper Carboniferous reference in a recent compilation (Bachtadse and Briden, 1990). By simply drawing a curve through these poles (Fig. 11), we obtain a rough African APWP, close to that of Feinberg et al. (1990). It appears first that the new Moscovian palaeopole issued from this study is in an intermediate position (Fig. 11) between the Upper Namurian–Lower Moscovian poles of the Hassi-Bachir and Ain Ech Chebbi formations (Daly and Irving, 1983) on the one hand, and the Stephano–Autunian pole of Abadla levels (Morel et al., 1981) on the other hand. Secondly, this new African APWP derived from African data (except the Dwyka poles) agrees particularly well with the migration trend of glacial centres during the Paleozoic (Caputo and Crowell, 1985) and with the lithological indicators of climate (Scotese and Barrett, 1990).

However, we must point out that the supposed Upper Carboniferous pole in the Dwyka series from central Africa is close to the Early Carboniferous pole of Ben Zireg (Aifa et al., 1989). This discrepancy might be due to an imprecise dating of the sampled part of the Dwyka formation, as assumed by McElhinny and Opdyke in their original paper. In fact, the Dwyka series has usually been regarded as being Upper Carboniferous in age, but the fossil evidence, well summarized by Du Toit (1954), does not allow us to exclude an age prior to Upper Carboniferous for the older part of the Dwyka series (Plumsteadt, 1967; Rust, 1973, Fabre, 1988).

Geodynamical implications

The general trend of the African APWP (Fig. 11) clearly shows that the main *latitudinal* displacement of Africa (Fig. 12) is a northwards drift from Famennian to Tournaisian. Comparison of this Tournaisian pole with the European APWP (Van der Voo, 1990) shows that no very wide ocean (Van der Voo, 1988; Kent and Van der Voo, 1990) separated southern Europe and northern Africa at this time (this is in good agreement with sedimentary and faunal evidence—Robardet et al., 1990). Later, probably until the Namurian, the general trend of the African APWP implies a clockwise rotation, of about 25° (Aifa et al., 1990), with the centre of rotation close to the northeastern border of Africa (about the present location of the Red Sea shoreline in southern Egypt). After the Moscovian period (pole 'a' in Fig. 11), the main latitudinal displacement of Africa is again a northwards drift. The first change in the drift in the Early Carboniferous could be explained by the "blocking" of Gondwana during the main collision against the Laurussian plate, dating the beginning of the Hercyno–Appalachian orogeny. The second

TABLE 2

Upper Devonian to Permian palaeomagnetic pole positions for stable Africa

Formation	Age	Lat. (° S)	Long. (° E)	Reference
Ben Zireg	Famennian	19.2	19.8	Aifa et al., 1989
Ben Zireg	Tournaisian	25.3	21.1	Aifa et al., 1989
Dwyka	Lower Carboniferous?	26.5	26.5	McElhinny and Opdyke, 1968
Hassi Bachir	U. Namurian–L. Moscovian	26.8	56.6	Daly and Irving, 1983
Ain Ech Chebbi	Lower Moscovian	22.9	51.8	Daly and Irving, 1983
El Adeb Larache	Moscovian	28.7	55.9	This study
Abadla	Stephano–Autunian	29	60	Morel et al., 1981
El Adeb Larache	Stephano–Autunian	38.5	57.5	This study
Hassi Bachir	Permian overprint	35.5	60.0	Daly and Irving, 1983
Beni-Abbes	Permian overprint	49.5	42.2	Aifa, 1987
El Adeb Larache	Permian overprint	41.2	65.9	This study

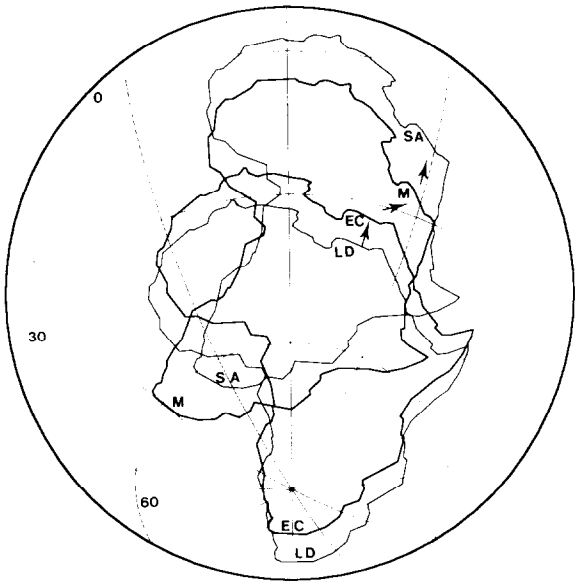


Fig. 12. Latitudinal displacement of Africa according to data in Fig. 11: Late Devonian (*LD*), Early Carboniferous (*EC*), Moscovian (*M*) and Stephano–Autunian (*SA*) periods. Centre of rotation during Carboniferous for latitudinal displacement (invariant latitude spot) here arbitrarily chosen as invariant longitude spot.

change could date the last phases of the Pangea formation. The Gondwana rotation between these two periods can be explained by the fact that the collision affected at first only a part of Gondwana and Laurussia, thus implying a blocking of the northwards drift only for this first main collision area. Such a displacement is compatible with the Pangea A2 (Van der Voo, 1981) or B (Irving, 1977) hypotheses. Our model also explains one of the major objections against the Pangea A2 assumption, i.e. the lack, in Permian times, of a continent south of central and eastern Europe which was, however, a continental collision area during the Hercynian orogeny (see Matte, 1986). In fact, as in Matte's (1986) model, collision could have begun mainly from Morocco to Bohemia (even perhaps to the Caucasus), and only later (Hatcher and Odom, 1980; Lécorché et al., 1989) been extended, because of the Gondwana rotation, to Mauritania–southern Appalachians with the opening of the Palaeo-Tethys south of eastern Europe; such an assumption also does not imply the dextral mega-movement during Permian times between Laurussia and Gondwana, assumed in the Pangea B hypothesis and presently not well supported by substantial geological evidence.

Conclusion

The El Adeb Larache levels have presented several favourable factors for a palaeomagnetic study. This well-dated Moscovian formation is situated in a stable area and its outcrop conditions here allowed a comprehensive sampling covering all its stratigraphic levels.

The majority of these samples allows to infer a new Moscovian pole. It represents an important contribution to the APWP of Africa and pinpoints the previously controversial location of this continent during the Upper Carboniferous. Our interpretation carries the inference that the Dwyka "varves" pole (McElhinny and Opdyke, 1968) is of a Lower Carboniferous rather than Upper Carboniferous age, as concluded in the original study.

Comparison with Upper Devonian–Lower Carboniferous data allows us to show an important clockwise rotation of Africa, thus explaining the evolution of the main collision area between Gondwana and Laurussia during the Carboniferous.

Acknowledgements

We are very grateful to the Algerian DPRS, the French Foreign Office and SONATRACH (especially Dr. Attar and Dr. Hamel, and the geological staff of In Amenas) which greatly supported this work, to M. Le Goff, H. Harchaoui and R. Bendjelloul for assistance in the field, to anonymous reviewers for constructive comments, and to T. Bedad for help with the manuscript.

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